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Subject: Computer Vision Laboratory

Experiment - 9

Aim: To classify images using BoVW model with LBP and HOG features.

Software Tools Required: VS Code, Python 3.12

Theory:

1. Introduction

Image classification aims to categorize an image into a predefined class based on its content. Traditional computer vision techniques (before deep learning became dominant) relied on hand-crafted features such as LBP and HOG to represent textures, shapes, and patterns in an image. However, these features by themselves may not generalize well for large datasets.

To overcome this, the Bag of Visual Words (BoVW) model—inspired by the Bag of Words model in text processing—was developed. It provides a robust way to represent an image as a distribution of "visual words" (clustered local features), making classification more effective.

2. Local Feature Descriptors

(a) Local Binary Patterns (LBP)

- LBP is a texture descriptor that captures local structural information.
- For each pixel, it compares the intensity with its neighboring pixels and encodes the result as a binary pattern.
- The final LBP feature vector is a histogram of these binary codes, representing the frequency of different texture patterns.
- Advantages: Simple, efficient, rotation-invariant (with modifications), and good for texture-based recognition tasks.

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(b) Histogram of Oriented Gradients (HOG)

- HOG is a shape/edge descriptor that captures the distribution of gradient orientations in localized portions of an image.

- The image is divided into cells, gradients are computed, and histograms of orientations are built.
- Normalization across blocks ensures invariance to illumination and contrast.
- Advantages: Captures edge and contour information, robust to illumination changes, effective in object detection (famously used in pedestrian detection).

3. Bag of Visual Words (BoVW) Model

The BoVW model adapts the idea of word histograms from Natural Language Processing (NLP) into computer vision:

1. **Feature Extraction:** Extract LBP and HOG descriptors from local patches in all images.
2. **Codebook Generation (Dictionary Learning):**
   - Apply a clustering algorithm (e.g., k-means) to group similar feature vectors into clusters.
   - Each cluster center is considered a "visual word".
   - The collection of visual words forms the visual vocabulary or dictionary.
3. **Feature Encoding:**
   - Each local descriptor in an image is assigned to the nearest visual word.
   - The image is then represented as a histogram of visual word occurrences (frequency count).

- This histogram is a fixed-length vector, independent of the original image size.

4. **Classification:**
   - The BoVW histograms are used as input features to a classifier (e.g., Support Vector Machine (SVM), Random Forest, k-NN).
   - The classifier learns decision boundaries and assigns test images to their respective categories.

4. Combining LBP and HOG in BoVW

- **Why combine?**
  - LBP captures fine texture and local micro-patterns. o HOG captures global shape and edge structure.
  - Together, they provide a complementary representation of the image.
- **Approach:**
  - Extract both LBP and HOG features from image patches.
  - Concatenate the feature descriptors before clustering (early fusion), or generate separate vocabularies and concatenate histograms later (late fusion).
  - Encode features into BoVW histograms.
  - Use these histograms as input to a classifier.

5. Advantages of BoVW with LBP and HOG

- Works well on small to medium-sized datasets without requiring deep learning.
- Robust against variations in scale, rotation, and illumination.
- Provides a compact and discriminative feature representation.
- Combining LBP and HOG increases discriminative power since they cover both texture and shape information.

6. Limitations

- Loses spatial information (histograms ignore the exact location of patterns).
- Requires careful selection of vocabulary size (too small → poor representation, too large → overfitting).
- Computational cost increases with larger dictionaries and feature sets.

Flowchart

Part-1

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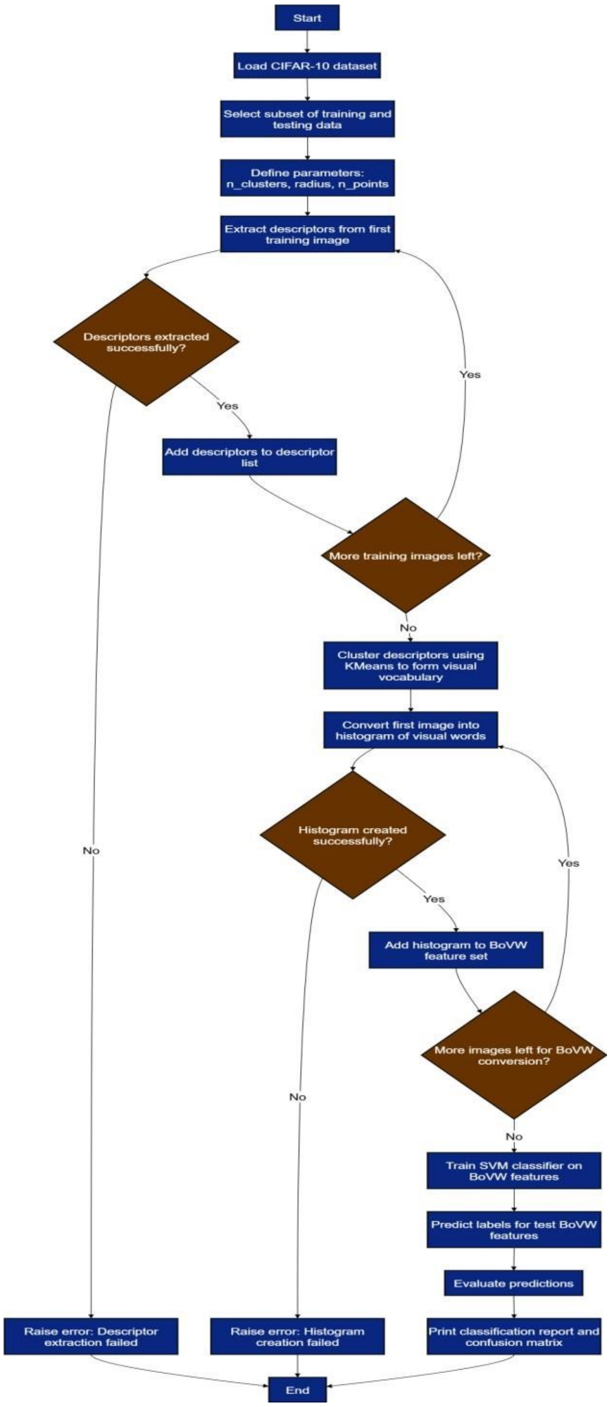

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Program with Output:

Program:

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```python
import numpy as np
from sklearn.cluster import KMeans
from sklearn.svm import SVC
from sklearn.metrics import classification_report, confusion_matr
from skimage.feature import local_binary_pattern, hog
from tensorflow.keras.datasets import cifar10

# Parameters
n_clusters = 50  # Size of visual vocabulary
radius = 1
n_points = 8 * radius

# Load CIFAR-10 data
(X_train, y_train), (X_test, y_test) = cifar10.load_data()
X_train = X_train[:500]  # Use a subset for speed
y_train = y_train[:500].flatten()
X_test = X_test[:100]
y_test = y_test[:100].flatten()

# Step 1: Extract descriptors (LBP + HOG) from patches
def extract_descriptors(img):
    img_gray = np.mean(img, axis=2).astype(np.uint8)
    # LBP
    lbp = local_binary_pattern(img_gray, n_points, radius, method
    lbp_desc = lbp.flatten()
    # HOG
```

```python
    # HOG
    hog_desc = hog(img_gray, pixels_per_cell=(8, 8), cells_per_bl
    # Combine
    return np.concatenate([lbp_desc, hog_desc])

# Build descriptor list for all images
desc_list = []
for img in X_train:
    desc = extract_descriptors(img)
    desc_list.append(desc)
desc_list = np.array(desc_list)

# Step 2: Cluster descriptors to form visual vocabulary
kmeans = KMeans(n_clusters=n_clusters, random_state=0)
kmeans.fit(desc_list)

# Step 3: Represent each image as histogram of visual words
def img_to_bovw(img):
    desc = extract_descriptors(img)
    word = kmeans.predict([desc])[0]
    hist = np.zeros(n_clusters)
    hist[word] += 1
    return hist
```

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```python
# Step 3: Represent each image as histogram of visual words
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    desc = extract_descriptors(img)
    word = kmeans.predict([desc])[0]
    hist = np.zeros(n_clusters)
    hist[word] += 1
    return hist

X_train_bovw = np.array([img_to_bovw(img) for img in X_train])
X_test_bovw = np.array([img_to_bovw(img) for img in X_test])

# Step 4: Train SVM classifier
clf = SVC(kernel='linear')
clf.fit(X_train_bovw, y_train)

# Step 5: Evaluate
y_pred = clf.predict(X_test_bovw)
print("Classification Report:\n", classification_report(y_test, y
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Output:

```
Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        10
           1       0.07      1.00      0.12         6
           2       0.00      0.00      0.00         8
           3       0.00      0.00      0.00        10
           4       0.00      0.00      0.00         7
           5       0.00      0.00      0.00         8
           6       0.00      0.00      0.00        16
           7       0.00      0.00      0.00        11
           8       0.12      0.08      0.10        13
           9       0.00      0.00      0.00        11

    accuracy                           0.07       100
   macro avg       0.02      0.11      0.02       100
weighted avg       0.02      0.07      0.02       100
```

```
Confusion Matrix:
 [[ 0  8  0  0  0  0  0  0  2  0]
 [ 0  6  0  0  0  0  0  0  0  0]
 [ 0  7  0  0  0  0  0  0  1  0]
 [ 0 10  0  0  0  0  0  0  0  0]
 [ 0  7  0  0  0  0  0  0  0  0]
 [ 0  8  0  0  0  0  0  0  0  0]
 [ 0 10  0  0  0  0  0  0  0  0]
 [ 0  7  0  0  0  0  0  0  0  0]
 [ 0  8  0  0  0  0  0  0  0  0]
 [ 0  7  0  0  0  0  0  0  0  0]
 [ 0  8  0  0  0  0  0  0  0  0]
 [ 0  8  0  0  0  0  0  0  0  0]
 [ 0 16  0  0  0  0  0  0  0  0]
 [ 0  8  0  0  0  0  0  0  3  0]
 [ 0 12  0  0  0  0  0  0  1  0]
 [ 0  8  0  0  0  0  0  0  3  0]
 [ 0 12  0  0  0  0  0  0  1  0]
 [ 0 12  0  0  0  0  0  0  1  0]
 [ 0 10  0  0  0  0  0  0  1  0]]
```

Conclusion/Inferences:

The BoVW model with LBP and HOG features effectively combines texture and shape information for image classification. LBP captures fine local patterns, while HOG extracts edge structures, and BoVW converts them into a robust histogram representation for classification. This approach is efficient and reliable for small to medium-scale datasets, though less effective on very large or complex datasets compared to deep learning methods.